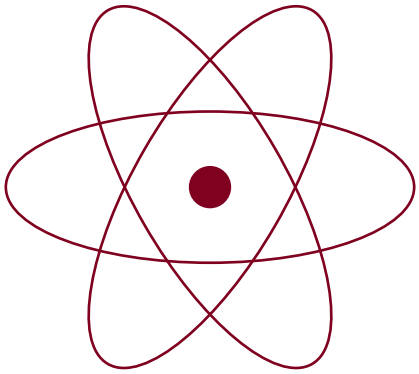
SPhOH

SAUDI PHYSICS OLYMPIAD HUB



Kinematics I

Olympiad Problem Set

Olympiad Training • Mechanics

SPhOH Team

SECOND EDITION • 2026

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About This Problem Set

This problem set is part of the training resources of the **Saudi Physics Olympiad Hub (SPhOH)**.

It is designed primarily for students preparing for the Saudi Physics Olympiad and related advanced physics competitions. The purpose of the handout is to provide a focused and substantial collection of problems for deliberate problem-solving practice.

The handout is **problem-centered**. Explanations are intentionally brief and are included only where they help establish the ideas or techniques needed for the problems.

How to Use the Problem Set

Students are encouraged to attempt each problem independently before looking at its hint. A hint should be treated as a way to overcome a specific obstacle, not as a substitute for solving the problem.

For difficult problems, several progressive hints may be provided. Try to use the least amount of assistance necessary before returning to the problem.

Difficulty Levels

The normal problems are divided into three training levels:

- **Level I** — Easy and direct substitution problems.
- **Level II** — Problems requiring combinations of ideas and more substantial reasoning.
- **Level III** — National Olympiad level problems.

Find more at: `https://devfhd.github.io/SPhOH/.`

CHAPTER 1

Kinematics I

Topic Note: Kinematic Relations
Kinematics describes motion without considering the forces causing it. Key relationships include:

$$v_{\text{avg}} = \frac{\Delta x}{\Delta t}, \quad a = \frac{\Delta v}{\Delta t}$$

For uniform acceleration a, where v_0 and v represent initial and final velocities while x and Δx represent position and change in position:

$$v = v_0 + at, \quad x = x_0 + v_0 t + \frac{1}{2}at^2, \quad v^2 = v_0^2 + 2a\Delta x$$

In 2D projectile motion, horizontal and vertical motions are independent:

$$x(t) = (v_0 \cos\theta)t, \quad y(t) = (v_0 \sin\theta)t - \frac{1}{2}gt^2$$

The horizontal range R and maximum height H on level ground are given by:

$$R = \frac{v_0^2 \sin(2\theta)}{g}, \quad H = \frac{v_0^2 \sin^2\theta}{2g}$$

1.1 1D Kinematics and Kinematic Equations

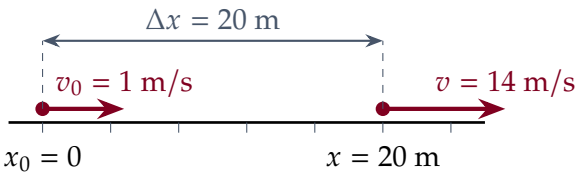
PROBLEM 1

LEVEL I

Acceleration along a Straight Line

Mohammed moves along a straight line with an initial velocity of 1 m/s. His velocity increases uniformly to 14 m/s while covering a distance of 20 m.

Determine the magnitude and direction of his acceleration.



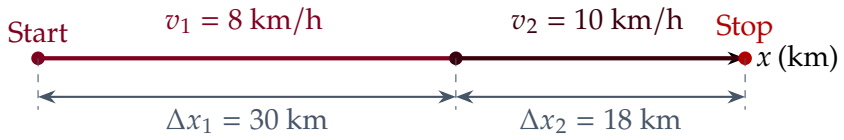
PROBLEM 2

LEVEL II

Multi-Stage Average Velocity

Hadi begins moving along a straight line at a constant speed of 8 1/h until he covers a distance of 30 . Immediately following this, he continues moving along the same line at a constant speed of 10 1/h for a distance of 18 , after which he comes to a complete stop.

Calculate the average speed for the entire trip (from the start of his motion until he stops).



PROBLEM 3

LEVEL II

Average Velocity over Split Distance and Time

A particle covers the first half of its total displacement with a constant velocity v_0. The second half of the distance is covered in two equal time intervals: moving with a constant velocity v_1 for the first half of that time, and with velocity v_2 for the remaining half of the time.

Calculate the average velocity v_{avg} of the particle averaged over the entire duration of its motion.

PROBLEM 4

LEVEL II

Motorboat and Floating Raft

A motorboat traveling downstream along a river passes a floating raft at point A. The boat continues downstream for a time interval $\tau = 60$ min, then turns around and travels upstream at the same power setting (constant speed relative to the water). The boat meets the raft again at a distance $l = 6.0$ downstream from point A.

Assuming the river flow velocity remains uniform and constant, determine the speed of the river current.

PROBLEM 5

LEVEL II

2D Motion under Constant Acceleration

A particle begins moving from an initial position vector $\vec{r}_0 = (3\hat{i} - 7\hat{j})$ m with an initial velocity $\vec{v}_0 = 2\hat{i}$ m/s. It moves with a constant acceleration of $\vec{a} = (\hat{i} + 3\hat{j})$ m/s^2.

(a) Will the particle cross the x-axis? If so, at what time t and at what x-coordinate?

(b) Will the particle cross the y-axis? If so, at what time t and at what y-coordinate?

1.2 Vector Displacement in 3D Space

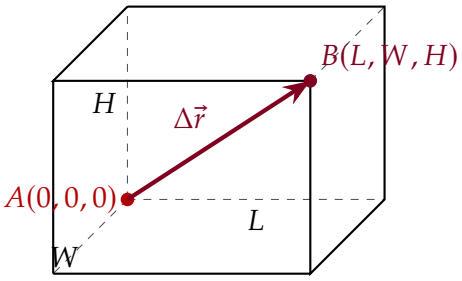
PROBLEM 6

LEVEL II

3D Box Corner-to-Corner Fly Displacement

A fly flies inside a rectangular room of dimensions $L \times W \times H$ from corner A (at the origin $(0, 0, 0)$) to the diagonally opposite corner B (at (L, W, H)).

Find the magnitude of the fly's displacement vector $\Delta\vec{r}$.



1.3 Graphical Analysis of Motion

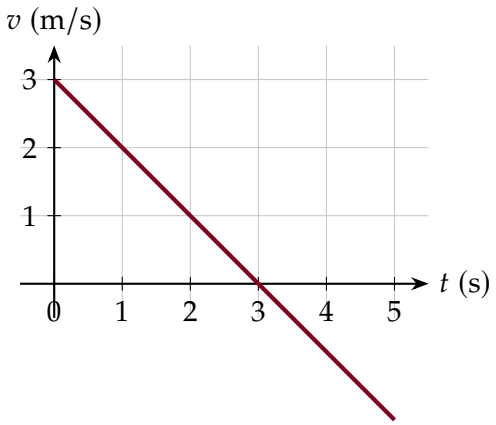
PROBLEM 7

LEVEL I

Position from Velocity–Time Graph

The graph below represents the velocity $v(t)$ of an object moving along a straight line as a function of time t. It is known that the position of the object at the instant when $v = 0$ is $x = 4$ m.

Determine the initial position of the object.



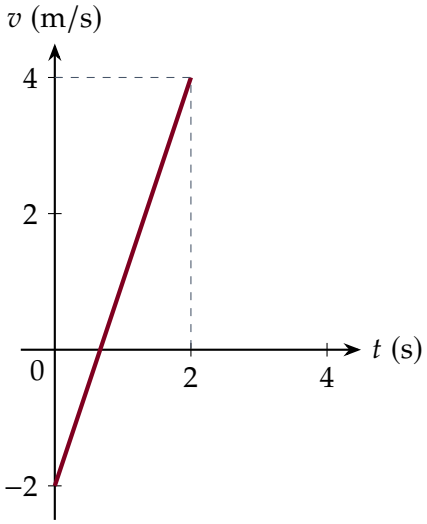
PROBLEM 8

LEVEL I

Acceleration from Velocity–Time Graph

The graph below depicts the velocity $v(t)$ of a particle moving along the x-axis over time.

Determine the instantaneous acceleration a of the particle at time $t = 0$.



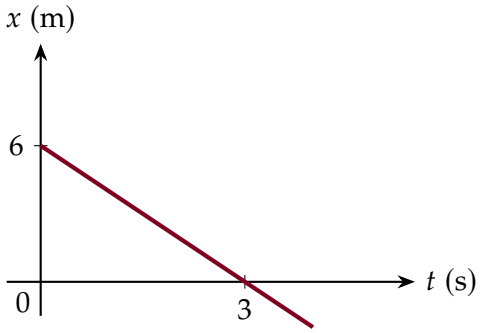
PROBLEM 9

LEVEL I

Instantaneous Velocity from Position–Time Graph

The position $x(t)$ of a particle moving along a straight line is plotted as a function of time t in the figure below.

Determine the instantaneous velocity v of the object at $t = 0$.



1.4 Free Fall and Vertical Motion under Gravity

PROBLEM 10

LEVEL III

Continuous Drop of Falling Balls

(a) A ball is released in air from rest. After time t_0, it falls down by distance is h. How much displacement would it take in the second t_0 interval? (Assume that it did not reach the ground yet)

(b) Another ball is released from the ground with some initial speed. After time t_0, its height is h. What is its height After the second t_0 interval ? (Assume that its velocity did not reach 0 yet)

(c) A child standing on the roof of a residential building releases (drops from rest) tennis balls vertically downward at regular time intervals. The rate of dropping is adjusted such that at the exact instant a new ball is released, the previous ball is at some intermediate position in the air, and the ball before that has just struck the ground. At the instant a ball is released by the child, what is the height above the ground of the ball currently in the air? The height of the building is H= 12 m.

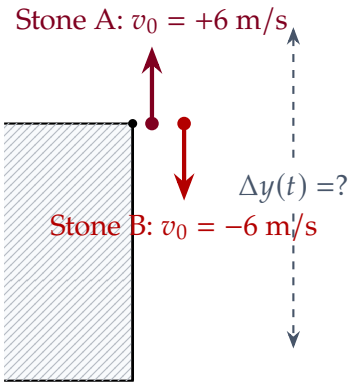
PROBLEM 11

LEVEL II

Separation of Oppositely Thrown Stones

A man standing at the edge of a tall vertical cliff throws two identical stones with the same initial speed $v_0 = 6$ m/s: Stone A is thrown vertically upward, and Stone B is thrown vertically downward.

Assuming neither stone has hit the ground below, what is the distance between the two stones after a time $t = 3$ s? (Take $g = 9.8$ m/s^2)



PROBLEM 12

LEVEL II

Collision of Two Counter-Moving Balls

Two balls are thrown simultaneously along the same vertical line. Ball 1 is launched from the ground vertically upward with an initial speed $v_1 = 20$ m/s. Ball 2 is released simultaneously from a height $H = 150$ m above the ground, thrown vertically downward with an initial speed $v_2 = 30$ m/s. (Take $g = 10$ m/s^2)

(a) Calculate the time t_{coll} elapsed until the two balls collide, and determine the height y_{coll} above the ground at which the collision occurs.

(b) Which event occurs first: Ball 1 reaching its maximum height, or the collision between the two balls?

(c) Determine the distance traveled by the first ball.

1.5 Projectile Motion

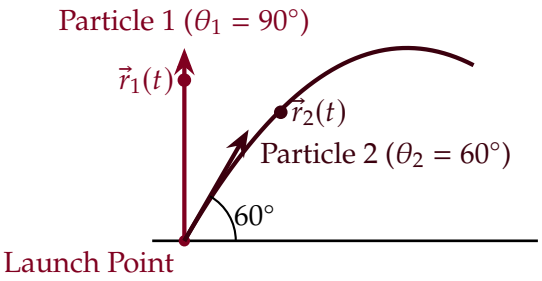
PROBLEM 13

LEVEL II

Diverging Projectiles Distance

Two particles are launched simultaneously from the exact same location with an identical initial speed $v_0 = 25$ m/s. The first particle is projected vertically upward ($\theta_1 = 90°$), while the second particle is launched at an angle $\theta_2 = 60°$ above the horizontal.

Ignoring air resistance, calculate the distance between the two particles at time $t = 1.70$ s after launch.



PROBLEM 14

LEVEL II

Direct Aim at a Vertical Wall

A projectile is launched from ground level at an angle $\theta = 30°$ above the horizontal with an initial speed $v_0 = 40$ m/s, aimed directly at a specific target located on a vertical wall at a horizontal distance $d = 70$ m.

Neglecting air resistance, calculate the vertical deflection from the target (i.e., the vertical distance between the intended target point and the actual impact point on the wall).

PROBLEM 15

LEVEL II

Speed versus Height Graph Analysis

The figure shows the speed v versus height y of a ball tossed directly upward, along a y axis[cite: 4]. Distance d is 0.40 m[cite: 4]. The speed at height y_A is v_A[cite: 4]. The speed at height y_B is $\frac{1}{3}v_A$[cite: 4].

What is speed v_A[cite: 4]?

PROBLEM 16

LEVEL II

Depth of a Water Well from Sound

A man drops a rock into a well[cite: 5].

(a) The man hears the sound of the splash 2.40 s after he releases the rock from rest[cite: 5]. The speed of sound in air (at the ambient temperature) is 336 m/s[cite: 5]. How far below the top of the well is the surface of the water[cite: 5]?

(b) **What If?** If the travel time for the sound is ignored, what percentage error is introduced when the depth of the well is calculated[cite: 5]?

PROBLEM 17

LEVEL II

Dive-Bomber Release Angle

A dive-bomber has a velocity of 280 m/s at an angle θ below the horizontal[cite: 2]. When the altitude of the aircraft is 2.15 , it releases a bomb, which subsequently hits a target on the ground[cite: 2]. The magnitude of the displacement from the point of release of the bomb to the target is 3.25 [cite: 2]. Find the angle θ[cite: 2].

PROBLEM 18

LEVEL II

Maximum Height to Range Ratio

A projectile is launched from level ground at an angle θ above the horizontal. Let R be its horizontal range and H be its maximum height.

(a) Derive an expression for the ratio H/R in terms of the launch angle θ.

(b) If a projectile is fired such that its horizontal range is equal to two times its maximum height ($R = 2H$), find the launch angle θ.

PROBLEM 19

LEVEL II

Fixed Initial Speed with Variable Launch Angle

A cannon fires projectiles at a fixed initial speed v_0, but the launch angle θ can be varied. The maximum possible horizontal range achievable by the cannon is L.

Determine the horizontal range of a projectile fired at an angle $\theta = 30°$ above the horizontal in terms of L.

PROBLEM 20

LEVEL II

Level-Ground Projectile with Range 60m

A projectile is launched from level ground at an initial angle $\theta = 45°$ and achieves a horizontal range $R = 60$ m. Neglecting air resistance:

(a) Determine the initial speed v_0 of the projectile.

(b) Determine the total time of flight T.

(c) How would the answers to (a) and (b) change if the range remained $R = 60$ m, but the launch angle were increased to $\theta > 45°$? Explain physically.

PROBLEM 21

LEVEL II

Impact Velocity and Vector Direction

A particle is launched from level ground with an initial velocity $\vec{v}_i$ at an angle θ_i above the positive x-axis. When it returns to ground level, its final velocity vector is $\vec{v}_f$.

Expressed in terms of v_i and θ_i:

(a) Find the angle between the initial velocity vector $\vec{v}_i$ and the final velocity vector $\vec{v}_f$.

(b) Express the sum of velocity vectors $\vec{v}_i + \vec{v}_f$ in unit vector notation $(\hat{i}, \hat{j})$.

PROBLEM 22

LEVEL III

Ramp Collision Sequence

A ball is thrown so that it hits a stationary ramp and then bounces up the ramp[cite: 1]. Immediately before the first collision between the ball and the ramp, the ball has a velocity v_0 that is horizontal[cite: 1]. Assume that all collisions are perfectly elastic, that there is no friction between the ramp and the ball, and that air resistance is negligible[cite: 1]. At the instant that the ball makes its $(N+1)^{\text{st}}$ collision with the ramp, the ball is moving perpendicular to the ramp[cite: 1]. Express your answers in terms of N, v_0, and g[cite: 1].

(a) Find the angle that the ramp makes with the horizontal[cite: 1].

(b) What is the distance, measured along the ramp, between the point at which the first collision occurs and the point at which the $(N+1)^{\text{st}}$ collision occurs[cite: 1]?

PROBLEM 23

LEVEL III

Tennis Ball Launcher on an Accelerating Train

A tennis ball launcher is placed on the floor at the front end of a train that has a uniform forward acceleration of 2.00 m/s^2[cite: 3]. The launcher projects a ball at an initial speed of 25.0 m/s with respect to the train toward its rear[cite: 3]. The ball achieves a maximum height of 10.0 m[cite: 3]. Ignore air resistance[cite: 3].

(a) Find θ_0, the angle of launch with respect to the floor[cite: 3].

(b) Find how far from the front end of the train the ball lands[cite: 3].

PROBLEM 24

LEVEL III

Cliff-Top Projectiles at Opposite Angles

Two identical balls are launched from the edge of a cliff at height $h = 40$ m above level ground with the same initial speed $v_0 = 30$ m/s. Ball 1 is launched at an angle $\theta = 30°$ above the horizontal, while Ball 2 is launched at $\theta = 30°$ below the horizontal.

(a) Find the difference in time of flight $\Delta t = t_1 - t_2$ between the two balls hitting the ground.

(b) Find the horizontal distance between their impact points on the ground.

(c) Compare your answers for time of flight and horizontal distance with those of a projectile launched from level ground with speed $v_0 = 30$ m/s at $\theta = 30°$ above the horizontal. What do you observe?

PROBLEM 25

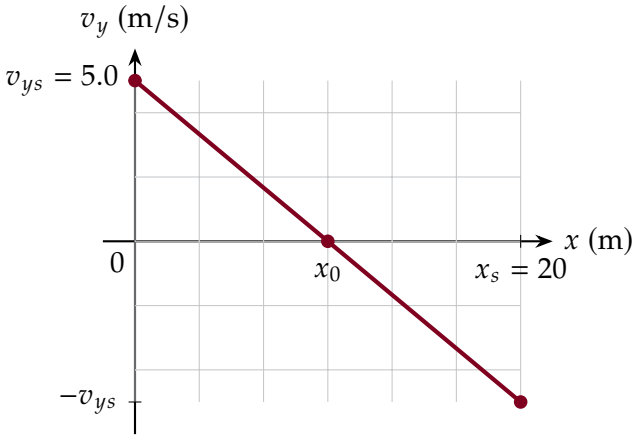
LEVEL III

Vertical Component of Impact Velocity against a Wall

A ball is shot from level ground toward a vertical wall located at a distance x from the launch point. The figure below shows the vertical velocity component v_y of the ball just as it reaches the wall, plotted as a function of the distance x.

The graph scaling is defined by $v_{ys} = 5.0$ m/s and $x_s = 20$ m.

Determine the launch angle θ_0 of the ball.



CHAPTER 2

Hints

The hints below correspond to the problems in the handout. They are provided separately so that students can work through the problem collection without immediately seeing the associated guidance.

Hint 1 | Problem 1 – Acceleration along a Straight Line

Use the kinematic equation relating velocity, acceleration, and displacement: $v^2 = v_0^2 + 2a\Delta x$. Solve for a.

Hint 2 | Problem 2 – Multi-Stage Average Velocity

Average speed is defined as total distance divided by total time: $v_{\text{avg}} = \frac{\Delta x_{\text{total}}}{\Delta t_{\text{total}}}$. Calculate the time spent in each stage using $\Delta t_i = \Delta x_i / v_i$.

Hint 3 | Problem 3 – Average Velocity over Split Distance and Time

First, find the effective average velocity v' for the second half of the journey. Since the time intervals are equal (t' and t'), $v' = \frac{v_1+v_2}{2}$. Then, combine the two equal-distance halves ($s/2$ and $s/2$) using the harmonic mean formula:

$$v_{\text{avg}} = \frac{2v_0v'}{v_0 + v'} = \frac{2v_0(v_1 + v_2)}{2v_0 + v_1 + v_2}$$

Hint 4 | Problem 4 – Motorboat and Floating Raft

Switch to the reference frame of the river water (in which the raft is stationary). In this frame, the water is at rest, so the motorboat moves away from the raft for a time τ and must take the exact same time τ to return to the raft! Therefore, the total time elapsed is 2τ, during which the raft floated a distance l. Thus, $v_{\text{flow}} = \frac{l}{2\tau}$.

Hint 5 | Problem 5 – 2D Motion under Constant Acceleration

Express the position vector as a function of time: $\vec{r}(t) = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2}\vec{a}t^2$. Separate into $x(t)$ and $y(t)$ components. Crossing the x-axis occurs when $y(t) = 0$, and crossing the y-axis occurs when $x(t) = 0$. Solve for $t > 0$.

Hint 6 | Problem 6 – 3D Box Corner-to-Corner Fly Displacement

The displacement vector is given by $\Delta\vec{r} = L\hat{i} + W\hat{j} + H\hat{k}$. The magnitude is found using the 3D Pythagorean theorem: $|\Delta\vec{r}| = \sqrt{L^2 + W^2 + H^2}$.

Hint 7 | Problem 7 – Position from Velocity–Time Graph

The displacement $\Delta x = x(t_2) - x(t_1)$ equals the area under the $v(t)$ curve between t_1 and t_2. Identify the time t when $v = 0$, compute the triangular area from $t = 0$ to that instant, and use $x(3) - x(0) =$ Area.

Hint 8 | Problem 8 – Acceleration from Velocity–Time Graph

Instantaneous acceleration is the slope of the velocity–time graph at that instant: $a(t) = \frac{\mathrm{d}v}{\mathrm{d}t}$. Find the slope of the line segment containing $t = 0$.

Hint 9 | Problem 9 – Instantaneous Velocity from Position–Time Graph

Instantaneous velocity is the slope of the position–time graph: $v(t) = \frac{\mathrm{d}x}{\mathrm{d}t}$. Calculate the slope of the line segment at $t = 0$.

Hint 10 | Problem 10 – Continuous Drop of Falling Balls

Let T be the time interval between consecutive drops. The ball striking the ground has fallen for total time $t_1 = 2T$, covering total height $H = \frac{1}{2}g(2T)^2 = 2gT^2$. The ball currently in the air has fallen for time $t_2 = T$, covering distance $y_2 = \frac{1}{2}gT^2$. Express y_2 as a fraction of H, then compute its height above the ground $h = H - y_2$.

Hint 11 | Problem 11 – Separation of Oppositely Thrown Stones

Write the position equation for each stone using the same coordinate system (e.g., origin at cliff edge, upward positive):

$$y_A(t) = v_0 t - \frac{1}{2}gt^2, \quad y_B(t) = -v_0 t - \frac{1}{2}gt^2$$

Subtract the two positions to find the relative separation $\Delta y(t) = y_A(t) - y_B(t)$. Notice that gravity cancels out!

Hint 12 | Problem 12 – Collision of Two Counter-Moving Balls

For part (a), set up the position equations $y_1(t) = v_1 t - \frac{1}{2}gt^2$ and $y_2(t) = H - v_2 t - \frac{1}{2}gt^2$. Equate $y_1(t_{\text{coll}}) = y_2(t_{\text{coll}})$ to solve for t_{coll}. For part (b), calculate the time for Ball 1 to reach apex ($t_{\text{apex}} = v_1/g$) and compare it with t_{coll}.

Hint 13 | Problem 13 – Diverging Projectiles Distance

Write the relative velocity vector between the two particles: $\vec{v}_{\text{rel}} = \vec{v}_2 - \vec{v}_1$. Since both particles experience the same downward gravitational acceleration $\vec{g}$, their relative acceleration is zero ($\vec{a}_{\text{rel}} = 0$). Thus, the relative motion is uniform: $d(t) = |\vec{v}_{\text{rel}}| \cdot t$.

Hint 14 | Problem 14 – Direct Aim at a Vertical Wall

The intended target height along the line of sight is $y_{\text{target}} = d\tan\theta$. The time of flight to reach the wall is $t = \frac{d}{v_0\cos\theta}$. The actual impact height is $y_{\text{impact}} = (v_0\sin\theta)t - \frac{1}{2}gt^2$. The vertical deflection is $\Delta y = y_{\text{target}} - y_{\text{impact}} = \frac{1}{2}gt^2$.

Hint 15 | Problem 15 – Speed versus Height Graph Analysis

Use the 1D kinematic relationship $v^2 = v_0^2 - 2g\Delta y$ between two points[cite: 6]. Setting $v_B^2 = v_A^2 - 2gd$ with $v_B = \frac{1}{3}v_A$ and $d = 0.40\ \text{m}$ allows you to solve directly for v_A.

Hint 16 | Problem 16 – Depth of a Water Well from Sound

For part (a), set $t_{\text{total}} = t_{\text{fall}} + t_{\text{sound}} = 2.40\ \text{s}$, where $t_{\text{fall}} = \sqrt{\frac{2h}{g}}$ and $t_{\text{sound}} = \frac{h}{v_s}$. Solve the resulting quadratic equation for $\sqrt{h}$. For part (b), calculate the depth $h_{\text{approx}} = \frac{1}{2}gt_{\text{total}}^2$ assuming sound travels instantly, and compute the percentage error $\frac{|h_{\text{approx}}-h|}{h}\times 100\%$.

Hint 17 | Problem 17 – Dive-Bomber Release Angle

First calculate the horizontal displacement x using the total magnitude of displacement $d = \sqrt{x^2+y^2}$ with $y = 2.15\ \text{km}$ and $d = 3.25\ \text{km}$. Then express x and y as functions of time t and angle θ, eliminate t, and solve the trigonometric equation for θ.

Hint 18 | Problem 18 – Maximum Height to Range Ratio

Use the formulas $H = \frac{v_0^2\sin^2\theta}{2g}$ and $R = \frac{v_0^2\sin(2\theta)}{g} = \frac{2v_0^2\sin\theta\cos\theta}{g}$. Divide H by R to show that $\frac{H}{R} = \frac{1}{4}\tan\theta$. For part (b), set $\frac{H}{R} = \frac{1}{2}$.

Hint 19 | Problem 19 – Fixed Initial Speed with Variable Launch Angle

The maximum range L occurs at $\theta = 45°$, where $L = \frac{v_0^2}{g}$. Express the range for $\theta = 30°$ as $R(30°) = \frac{v_0^2\sin(60°)}{g}$ and substitute L.

Hint 20 | Problem 20 – Level-Ground Projectile with Range 60m

Use $R = \frac{v_0^2\sin(2\theta)}{g}$ to find v_0. The time of flight is $T = \frac{2v_0\sin\theta}{g}$. For part (c), if $\theta > 45°$, $\sin(2\theta) < 1$, so a higher initial speed v_0 is required to achieve the same range, which also increases T.

Hint 21 | Problem 21 – Impact Velocity and Vector Direction

On level ground, the horizontal velocity component remains constant ($\vec{v}_{fx} = \vec{v}_{ix} = v_i\cos\theta_i\hat{i}$), while the vertical component reverses direction ($\vec{v}_{fy} = -\vec{v}_{iy} = -v_i\sin\theta_i\hat{j}$). Thus $\vec{v}_i = v_i\cos\theta_i\hat{i} + v_i\sin\theta_i\hat{j}$ and $\vec{v}_f = v_i\cos\theta_i\hat{i} - v_i\sin\theta_i\hat{j}$.

Hint 22 | Problem 22 – Ramp Collision Sequence

Set up a coordinate system aligned with the incline (x-axis up the ramp, y-axis perpendicular to it). For part (a), analyze the constant accelerations $g\sin\theta$ and $g\cos\theta$ along both axes and use the condition that $v_x = 0$ at the $(N+1)^{\text{st}}$ collision. For part (b), sum the displacements along the ramp over all N hops.

Hint 23 | Problem 23 – Tennis Ball Launcher on an Accelerating Train

Work in the accelerated frame of reference of the train, where the ball experiences an effective horizontal fictitious acceleration $a_{\text{rel}} = 2.00\ \text{m/s}^2$ pushing it backward relative to the train. Use $v_{y0}^2 = 2gH_{\text{max}}$ to determine θ_0, and then calculate the relative landing distance using $x_{\text{rel}}(t) = v_{x0}t + \frac{1}{2}a_{\text{rel}}t^2$.

Hint 24 | Problem 24 – Cliff-Top Projectiles at Opposite Angles

Notice that Ball 1 rises, reaches its peak, and returns to the cliff height ($y = 0$) moving downward at speed v_0 at angle 30° below the horizontal—which is identical to the initial launch condition of Ball 2! Thus, the difference in flight time Δt is exactly the time Ball 1 spends executing level-ground projectile motion above the cliff top.

Hint 25 | Problem 25 – Vertical Component of Impact Velocity against a Wall

From the graph, at $x = 0$, $v_{y0} = v_{ys} = 5.0\ \text{m/s}$. The vertical velocity reaches zero ($v_y = 0$) at $x = 10\ \text{m}$, which corresponds to the peak of the trajectory where $x = \frac{R}{2}$. Use $v_y(t) = v_{y0} - gt = 0$ to find the time to peak $t_p = \frac{v_{y0}}{g}$, and then $x(t_p) = v_x t_p = 10\ \text{m}$ to solve for $v_x = v_0\cos\theta_0$. Finally, evaluate $\tan\theta_0 = \frac{v_{y0}}{v_x}$.